Day 4 Study Notes

Mon 16 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Semiconductors and logic gates (75 min) · Maths: Vectors (part 1) (75 min) · Chemistry: Coordination compounds (part 3: bonding, magnetism, colour) (60 min)
Exam	1 h 15 min	The Day 4 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Semiconductors and logic gates

About one question per paper, usually a logic gate, a diode circuit or a fact about doping. The syllabus covers diodes and logic gates only; transistors are not included.

1. Energy bands and types of semiconductor

- Conductors have overlapping bands. Insulators have an energy gap above 3 eV. Semiconductors have a gap below 3 eV: silicon 1.1 eV, germanium 0.7 eV.
- **Intrinsic** (pure): electron and hole concentrations are equal, $n_e = n_h = n_i$.
- **n-type:** doped with a pentavalent atom (P, As, Sb), the donor. Electrons are the majority carriers.
- **p-type:** doped with a trivalent atom (B, Al, Ga, In), the acceptor. Holes are the majority carriers.
- A doped crystal is still electrically neutral.
- At a given temperature $n_e\, n_h = n_i^2$, whatever the doping.
- Conductivity $\sigma = e(n_e\mu_e + n_h\mu_h)$. The resistance of a semiconductor falls as temperature rises.

> **Solved example 1.** Silicon with $n_i = 1.5 \times 10^{16}\ \text{m}^{-3}$ is doped so that $n_h = 4.5 \times 10^{22}\ \text{m}^{-3}$. Find n_e.
> $n_e = \dfrac{n_i^2}{n_h} = \dfrac{2.25 \times 10^{32}}{4.5 \times 10^{22}} = 5 \times 10^{9}\ \text{m}^{-3}$.

2. The p-n junction diode

- At the junction, carriers diffuse across and leave a **depletion region** with a **barrier potential**: about 0.7 V for silicon and 0.3 V for germanium.

	Forward bias	Reverse bias
Connection	p side to positive terminal	p side to negative terminal
Depletion region	narrows	widens
Barrier	decreases	increases
Current	milliamperes, once the voltage passes the knee value	microamperes, from minority carriers; nearly constant until breakdown
Ideal diode behaves as	a closed switch	an open switch

In circuit questions, if a barrier potential is given, subtract it: $I = \dfrac{V - V_{\text{barrier}}}{R}$. If the diode is called ideal, treat the drop as zero.

3. Rectifiers

- **Half-wave** (one diode): conducts in one half cycle. Output frequency = input frequency.
- **Full-wave** (two diodes with a centre tap, or four in a bridge): conducts in both half cycles. Output frequency = $2\times$ input frequency.
- A capacitor across the load smooths the output.

4. Special-purpose diodes

Device	Bias	What it does
Zener diode	reverse, in breakdown	keeps the voltage across it constant at V_Z; used as a voltage regulator
LED	forward	emits light of energy equal to the band gap; needs a gap of about 1.8 eV or more for visible light
Photodiode	reverse	reverse current increases with light intensity; used as a light detector
Solar cell	none	generates a voltage from light; its I–V curve lies in the fourth quadrant

Zener regulator. The Zener is in parallel with the load, with a series resistor R_S. Then $I_S = \dfrac{V_{\text{in}} - V_Z}{R_S}$, $I_L = \dfrac{V_Z}{R_L}$ and $I_Z = I_S - I_L$.

LED wavelength. $\lambda = \dfrac{hc}{E_g}$. With $hc = 1240$ eV nm, $\lambda(\text{nm}) = \dfrac{1240}{E_g(\text{eV})}$.

5. Logic gates

A	B	OR: A+B	AND: A·B	NOR	NAND
0	0	0	0	1	1
0	1	1	0	0	1
1	0	1	0	0	1
1	1	1	1	0	0

- NOT inverts: 0 becomes 1 and 1 becomes 0.
- NAND and NOR are universal gates: any gate can be built from either one alone. A NAND or NOR with its inputs joined is a NOT gate.
- De Morgan's rules: $\overline{A + B} = \bar{A} \cdot \bar{B}$ and $\overline{A \cdot B} = \bar{A} + \bar{B}$.
- So a NOR gate fed with inverted inputs is an AND gate, and a NAND gate fed with inverted inputs is an OR gate.

Traps. In a diode network, first decide for each diode whether it is forward or reverse biased, and remove the reverse-biased branches. For a full-wave rectifier the ripple frequency is doubled. For combinations of gates, write the Boolean expression gate by gate and simplify with De Morgan's rules instead of filling a full truth table.

Read in NCERT: Physics Class 12 Part 2, chapter "Semiconductor Electronics: Materials, Devices and Simple Circuits".

Mathematics · Vectors (part 1)

Vectors and 3D geometry together gave about three questions per paper in 2026. Today covers the basics and the dot product; tomorrow covers the cross product.

1. Basics

- For $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$: magnitude $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ and unit vector $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$.
- If A and B have position vectors $\vec{a}$ and $\vec{b}$, then $\vec{AB} = \vec{b} - \vec{a}$.
- Section formula: the point dividing AB internally in the ratio $m : n$ has position vector $\dfrac{m\vec{b} + n\vec{a}}{m+n}$. The midpoint is $\dfrac{\vec{a} + \vec{b}}{2}$.
- Two vectors are collinear (parallel) when one is a scalar multiple of the other: $\vec{a} = \lambda\vec{b}$, which means their components are proportional.
- Three points A, B, C are collinear when $\vec{AB}$ and $\vec{BC}$ are parallel.

2. The dot product

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$$

- $\vec{a} \cdot \vec{a} = |\vec{a}|^2$, and $\hat{i} \cdot \hat{i} = 1$, $\hat{i} \cdot \hat{j} = 0$.
- Perpendicular vectors: $\vec{a} \cdot \vec{b} = 0$.
- Angle: $\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$.
- Projection of $\vec{a}$ on $\vec{b}$: $\dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$. As a vector: $\dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\vec{b}$.
- Work done by a force: $W = \vec{F} \cdot \vec{d}$.

3. Identities that solve most questions

- $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\,\vec{a} \cdot \vec{b}$
- $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\,\vec{a} \cdot \vec{b}$
- If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then $\vec{a} \cdot \vec{b} = 0$ and the vectors are perpendicular.
- If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\dfrac{|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2}{2}$.

> **Solved example 1.** $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ with $|\vec{a}| = 3$, $|\vec{b}| = 5$, $|\vec{c}| = 7$. Find the angle between $\vec{a}$ and $\vec{b}$.
> $\vec{a} + \vec{b} = -\vec{c}$, so $|\vec{a} + \vec{b}|^2 = |\vec{c}|^2$: $9 + 25 + 2(3)(5)\cos\theta = 49$. Hence $\cos\theta = \dfrac{1}{2}$ and $\theta = 60^\circ$.

> **Solved example 2.** Find the projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.
> $\vec{a} \cdot \vec{b} = 2 + 6 + 2 = 10$ and $|\vec{b}| = \sqrt{6}$. Projection $= \dfrac{10}{\sqrt{6}}$.

Method for "given $\vec{a}+\vec{b}+\vec{c}=\vec{0}$" questions. Move one vector to the other side and square both sides. Squaring a vector equation means taking the dot product of each side with itself, which turns it into an equation in magnitudes and one angle.

Traps. The projection of $\vec{a}$ on $\vec{b}$ divides by $|\vec{b}|$, not $|\vec{a}|$. $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}$ does not mean $\vec{b}=\vec{c}$; it means $\vec{a}$ is perpendicular to $\vec{b}-\vec{c}$. In the section formula, the ratio number next to a point multiplies the other point's vector.

Read in NCERT: Mathematics Class 12 Part 2, chapter "Vector Algebra": up to and including the scalar (dot) product, with the solved examples.

Chemistry · Coordination compounds (part 3: bonding, magnetism, colour)

This part supplies the numerical-answer questions of the chapter: unpaired electrons, magnetic moment and crystal field stabilisation energy. Learn the spectrochemical series first, because every answer depends on whether the ligand is strong or weak.

1. Valence bond theory: hybridisation and shape

Coordination number	Hybridisation	Shape	Example
4	sp^3	tetrahedral	$[NiCl_4]^{2-}$, $[Ni(CO)_4]$
4	dsp^2	square planar	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_2Cl_2]$
6	d^2sp^3 (inner orbital, low spin)	octahedral	$[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{4-}$
6	sp^3d^2 (outer orbital, high spin)	octahedral	$[CoF_6]^{3-}$, $[Fe(H_2O)_6]^{3+}$

Strong-field ligands force the metal's d electrons to pair up, freeing inner d orbitals for bonding. Weak-field ligands leave them unpaired.

2. Magnetic moment

Spin-only formula: $\mu = \sqrt{n(n+2)}$ BM, where n is the number of unpaired electrons.

Unpaired electrons	1	2	3	4	5
μ in BM	1.73	2.83	3.87	4.90	5.92

No unpaired electrons means diamagnetic.

Complex	Metal ion	Ligand field	Unpaired electrons
$[Co(NH_3)_6]^{3+}$	Co^{3+}, d^6	strong	0
$[CoF_6]^{3-}$	Co^{3+}, d^6	weak	4
$[Fe(CN)_6]^{4-}$	Fe^{2+}, d^6	strong	0
$[Fe(CN)_6]^{3-}$	Fe^{3+}, d^5	strong	1
$[Fe(H_2O)_6]^{3+}$	Fe^{3+}, d^5	weak	5
$[Ni(CN)_4]^{2-}$	Ni^{2+}, d^8	strong, square planar	0
$[NiCl_4]^{2-}$	Ni^{2+}, d^8	weak, tetrahedral	2
$[Ni(CO)_4]$	Ni(0), d^{10} after pairing	strong, tetrahedral	0

3. Crystal field theory

- In an octahedral field the five d orbitals split into a lower set of three, t_{2g}, and an upper set of two, e_g. The gap is Δ_o. Each t_{2g} electron is stabilised by $0.4\Delta_o$; each e_g electron is raised by $0.6\Delta_o$.

- In a tetrahedral field the order is reversed (e below t_2) and the gap is smaller: $\Delta_t = \frac{4}{9}\Delta_o$. Tetrahedral complexes are therefore high spin.
- If Δ_o is larger than the pairing energy P, electrons pair in t_{2g} (low spin). If it is smaller, they spread out (high spin). The two cases differ only for d^4 to d^7.
- Crystal field stabilisation energy: $\mathrm{CFSE} = (-0.4\, n_{t_{2g}} + 0.6\, n_{e_g})\Delta_o$, plus pairing energy if asked.

Spectrochemical series, weak to strong field:

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < EDTA^{4-} < NH_3 < en < CN^- < CO$

> **Solved example.** Find the CFSE of $[Co(NH_3)_6]^{3+}$, ignoring pairing energy.
> Co^{3+} is d^6 and NH_3 is a strong-field ligand here, so the configuration is $t_{2g}^6 e_g^0$. CFSE $= 6(-0.4)\Delta_o = -2.4\Delta_o$. For high-spin d^6 ($t_{2g}^4 e_g^2$) it would be $-1.6 + 1.2 = -0.4\Delta_o$.

4. Colour

- Colour comes from an electron jumping from t_{2g} to e_g (a d–d transition) by absorbing visible light. The complex shows the complementary colour.
- $[Ti(H_2O)_6]^{3+}$ (d^1) absorbs blue-green light and looks violet.
- Ions with d^0 or d^{10} configuration (Sc^{3+}, Ti^{4+}, Zn^{2+}, Cu^+) have no d–d transition and are colourless.
- A stronger ligand gives a larger Δ_o, so light of higher energy and shorter wavelength is absorbed.

5. Bonding in metal carbonyls

The M–CO bond has two parts that strengthen each other (synergic bonding): a σ bond from the lone pair on carbon to the metal, and a π bond from a filled metal d orbital back into the empty π* orbital of CO. The back donation strengthens the M–C bond and weakens the C–O bond. Shapes: $[Ni(CO)_4]$ tetrahedral, $[Fe(CO)_5]$ trigonal bipyramidal, $[Cr(CO)_6]$ octahedral.

> **Traps.** Find the oxidation state before counting d electrons: remove the 4s electrons first. NH_3 is strong-field with Co^{3+} but $[Ni(NH_3)_6]^{2+}$ is still paramagnetic, because d^8 has two unpaired electrons in any octahedral field. In $[Ni(CO)_4]$ the 4s electrons move into 3d, giving d^{10}. Tetrahedral complexes are never low spin in this syllabus.

Read in NCERT: Chemistry Class 12 Part 1, chapter "Coordination Compounds": bonding in coordination compounds (valence bond theory, crystal field theory, colour) and bonding in metal carbonyls.